

EMCH 260

SOLID MECHANICS

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Bending

6

Chapter 6.3

- ✓ Determine bending deformation of a straight member
- ✓ Understand the flexure formula and moment of inertia
- ✓ Know how to calculate maximum normal stress in a member subjected to bending

PREVIOUS ON GRAPHICAL METHOD FOR SHEAR AND MOMENT DIAGRAMS

Regions of Distributed Load

$$\frac{dV}{dx} = w$$

Slope of shear diagram = distributed load intensity at each point

$$\frac{dM}{dx} = V$$

Slope of moment diagram = shear at each point

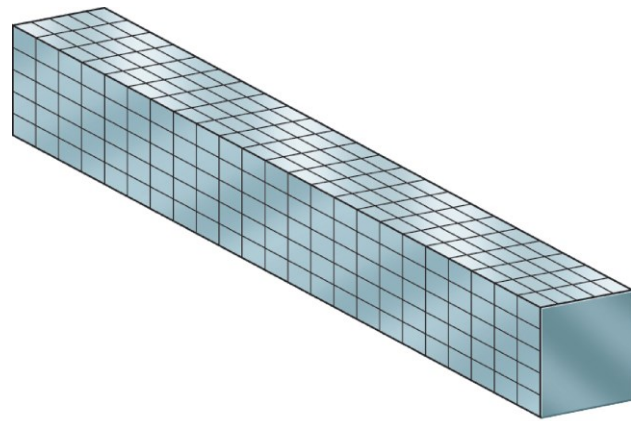
Regions of Concentrated Force and Moment

Upward (downward) F results in positive (negative) change in shear (ΔV), corresponding to the “jump” upward (downward) in shear diagram

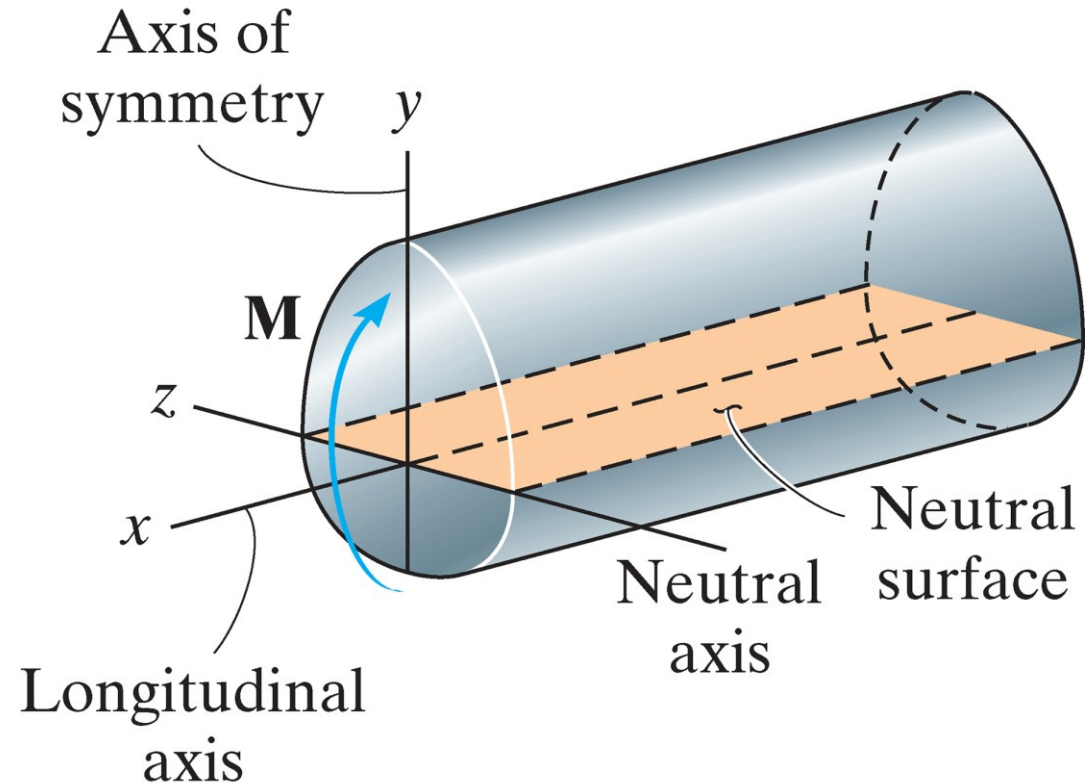
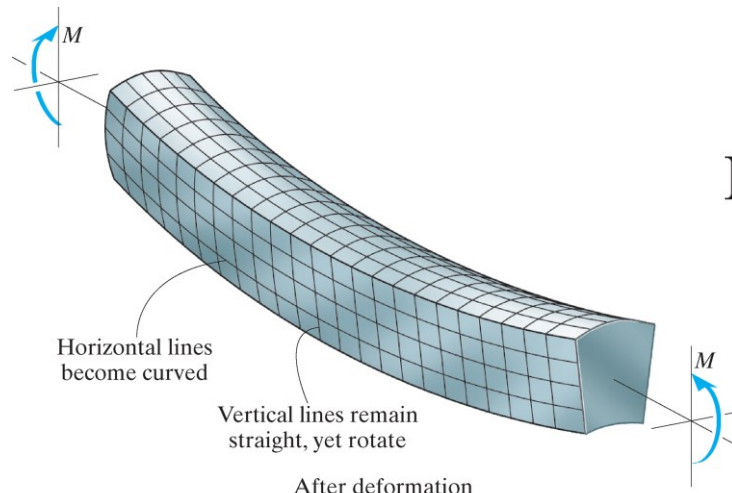
Clockwise (counterclockwise) M results in positive (negative) change in moment (ΔM), corresponding to the “jump” upward (downward) in moment diagram

BENDING DEFORMATION OF A STRAIGHT MEMBER

Neutral Surface & Neutral Axis

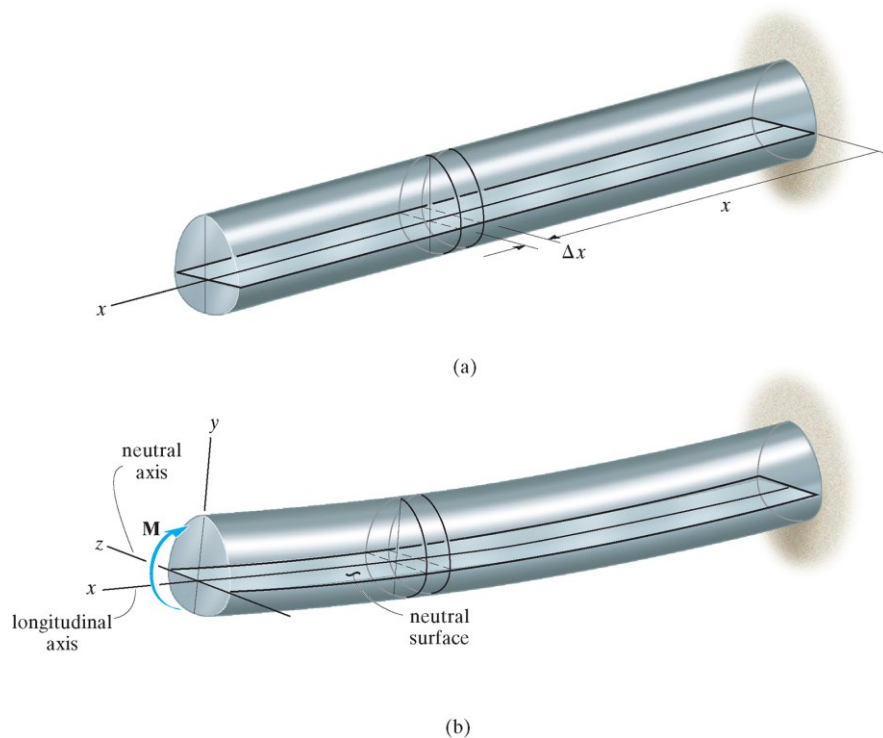


Before deformation



BENDING DEFORMATION OF A STRAIGHT MEMBER

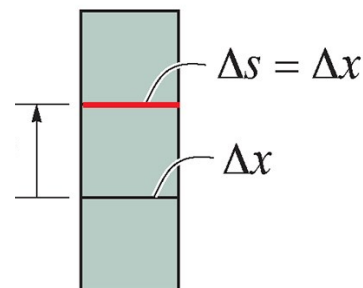
Normal strain along the longitudinal axis



$$\varepsilon = \lim_{\Delta s \rightarrow 0} \frac{\Delta s' - \Delta s}{\Delta s}$$

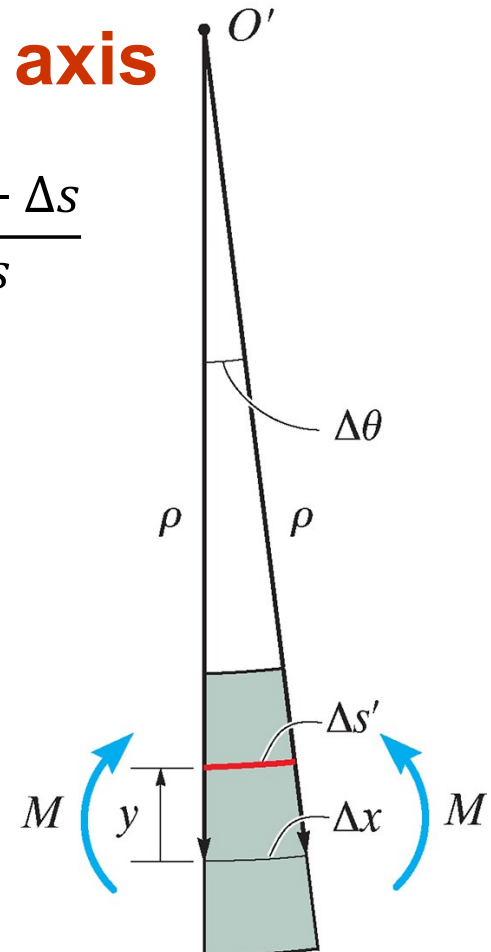
$$\Delta s = \rho \Delta \theta$$

$$\varepsilon = -\frac{y}{\rho}$$



Before deformation

(a)

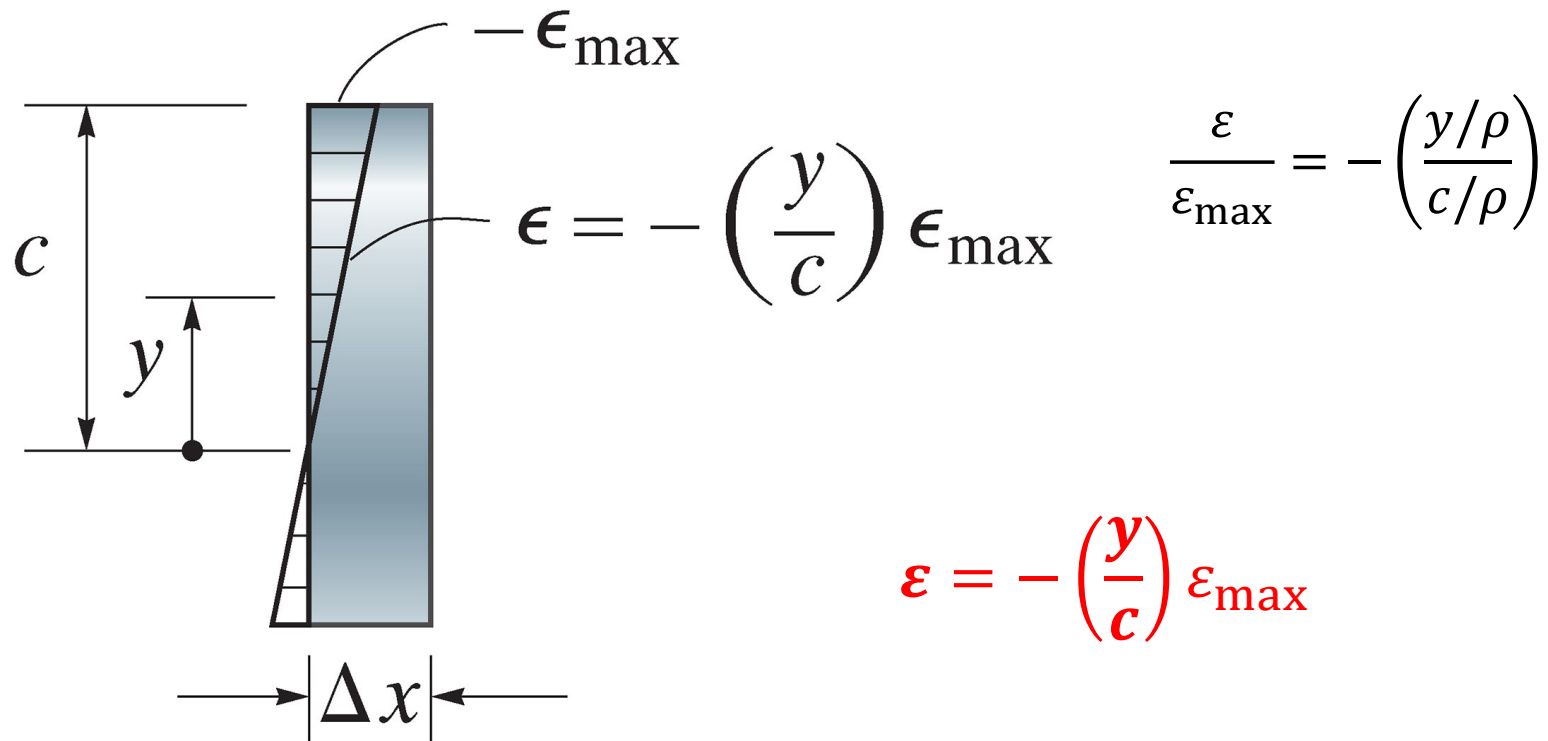


After deformation

(b)

Inner area is compressed ($\varepsilon < 0$)
Outer area is stretched ($\varepsilon > 0$)

NORMAL STRAIN DISTRIBUTION



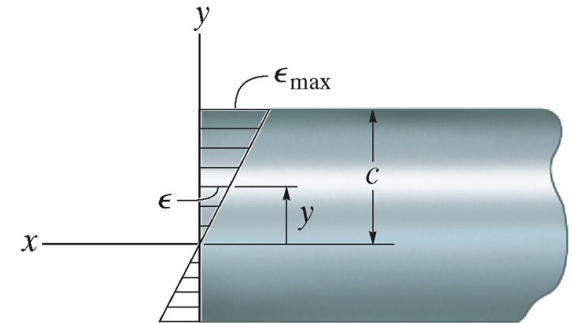
The maximum strain occurs at the outermost surface, located a distance of $y = c$ from the neutral axis.

NORMAL STRAIN DISTRIBUTION

- Assume that material behaves in a linear-elastic manner so that Hooke's law applies.
- A *linear variation of normal strain must* then be the consequence of a *linear variation in normal stress*
- Applying Hooke's law

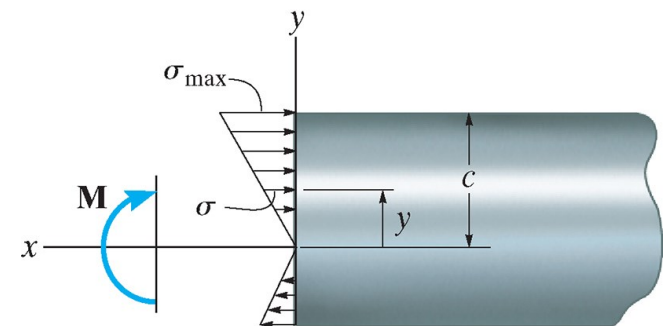
$$\sigma = -\left(\frac{y}{c}\right) \sigma_{\max}$$

Compressive stress ($\sigma < 0$) acts in the negative x -direction and occurs in the positive y region.



Normal strain variation
(profile view)

(a)



Bending stress variation
(profile view)

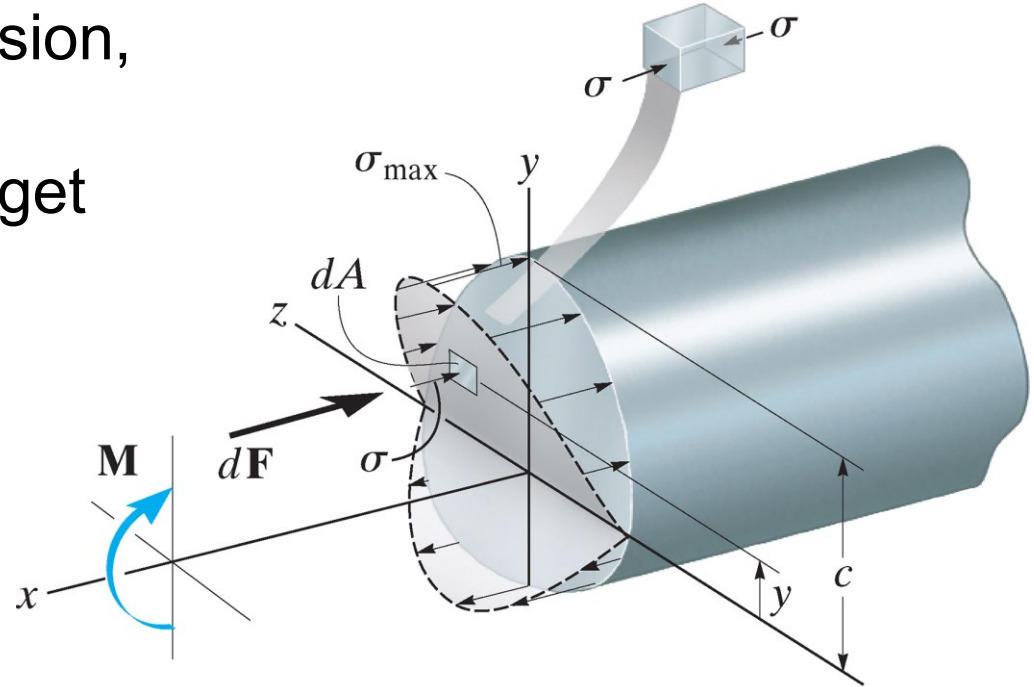
(b)

LOCATION OF NEUTRAL AXIS AND BENDING MOMENT

- By mathematical expression, equilibrium equations of moment and forces, we get

$$\int_A y dA = 0$$

$$M = \frac{\sigma_{\max}}{c} \int_A y^2 dA$$



- The integral represents the **moment of inertia** of x -sectional area, computed about the neutral axis. We symbolize its value as I .

MAXIMUM NORMAL STRESS

- Maximum normal stress can be solved and written as

$$\sigma_{\max} = \frac{Mc}{I} \quad I \equiv \int_A y^2 dA$$

$\sigma_{\max}$ = maximum normal stress in member, at a pt on x-sectional area farthest away from neutral axis

M = resultant internal moment, computed about neutral axis of x-section

I = moment of inertia of x-sectional area computed about neutral axis

c = perpendicular distance from neutral axis to a pt farthest away from neutral axis, where $\sigma_{\max}$ acts

FLEXURE FORMULA

- Normal stress at any intermediate distance y can be determined from

$$\sigma = -\frac{My}{I} \quad I \equiv \int_A y^2 dA$$

The negative sign indicates that, the *compressive* stress ($\sigma < 0$) occurs in the *positive* y region, while the *tensile* stress ($\sigma > 0$) occurs in the *negative* y region.

PROCEDURE FOR ANALYSIS

Step 1: Internal Moment

- Obtain moment **M** at the section (usually require shear and moment diagram to determine the maximum moment)
- Centroidal or neutral axis for x-section must be known since **M** is computed about this axis

Step 2: Section Property

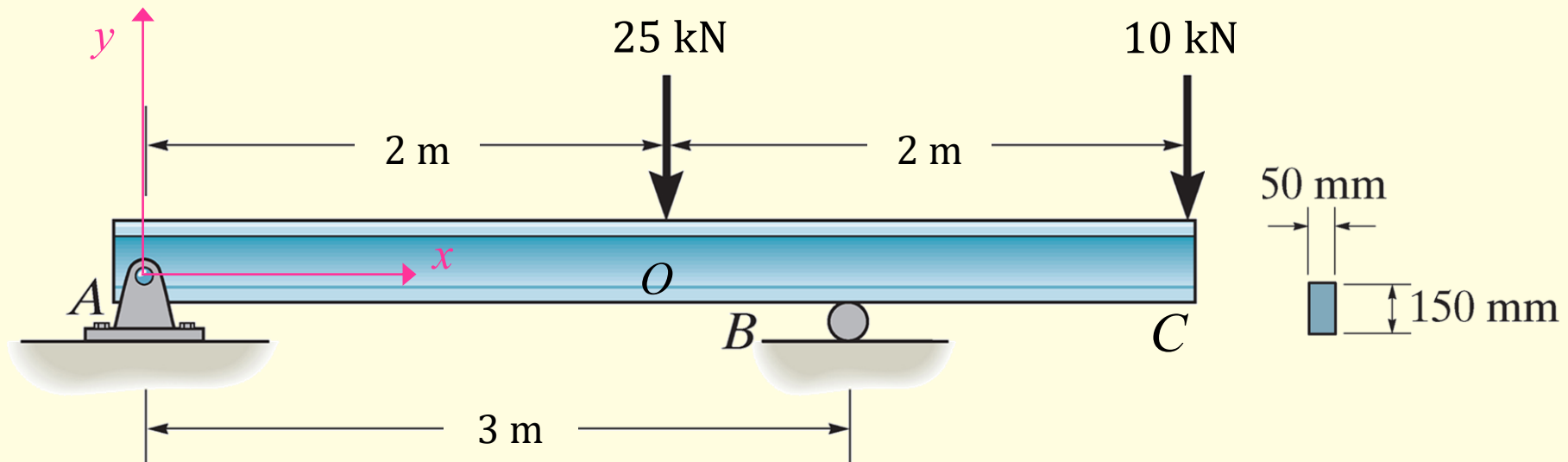
- Determine moment of inertia *I* of x-sectional area about the neutral axis
- Refer to the course book's inside front cover for the values of *I* for several common shapes (circular, triangular, rectangular)

Step 3: Normal Stress

- Specify distance *y*, measured perpendicular to neutral axis to the point where normal stress is to be determined
- Apply equation $\sigma = -\frac{My}{I}$, or if maximum bending stress is needed, use
$$\sigma_{\max} = \frac{Mc}{I}$$
- Be careful about the sign of stress

IN-CLASS EXAMPLE

The beam has a rectangular cross section. Determine the maximum bending stress in the beam if two external loadings are applied.



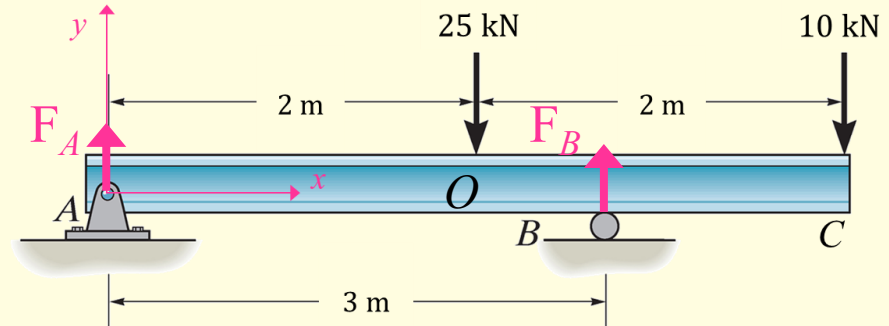
IN-CLASS EXAMPLE, cont'd

Solutions

Step 1: Support Reactions

Applying free-body diagram to the entire beam, the force equation of equilibrium gives:

$$\sum F_y = 0 \rightarrow F_A + F_B = 35 \text{ kN} \quad (1)$$



The moment equation of equilibrium with respect to point A gives:

$$\sum M_A = 0 \rightarrow F_B \times 3 = 25 \times 2 + 10 \times 4 \quad (2)$$

Solving Eqs. (1) and (2) gives:

$$F_A = 5 \text{ kN}$$

$$F_B = 30 \text{ kN}$$

IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Shear Diagram

a) Determine the shear at left end.

$$x = 0, \quad V = F_A = 5 \text{ kN}$$

b) Determine **V** diagram according to $\frac{dV}{dx} = w$ for AO region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow V = C_1$$

The constant C_1 can be determined by applying the “boundary” condition ($x = 0, V = 5 \text{ kN}$). Thus, $C_1 = 5 \text{ kN}$ and **$V = 5 \text{ kN}$ ($0 \leq x \leq 2 \text{ m}$)**.

c) At point O, due to the external concentrated force 25 kN (downward), the **V** decreases (downward jump) by 25 kN. Then, the **V** jumps from 5 kN to – 20 kN. See the shear diagram for details.

IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Shear Diagram (Cont'd)

d) Determine **V** diagram according to $\frac{dV}{dx} = w$
for *OB* region:

$$\frac{dV}{dx} = w = 0$$
$$\rightarrow V = C_2$$

The constant C_2 can be determined by the “**continuity**” condition at last point O after considering the downward jump.

Thus, $C_2 = -20$ kN and
 $V = -20$ kN $(2 \text{ m} \leq x \leq 3 \text{ m})$

e) At point *B*, due to the support reaction force F_B (upward), the **V** increases (upward jump) by $F_B = +30$ kN. Then, the **V** jumps from -20 kN to $+10$ kN.

IN-CLASS EXAMPLE, cont'd

Solutions

Step 2: Shear Diagram (Cont'd)

f) Determine **V** diagram according to $\frac{dV}{dx} = w$
for *BC* region:

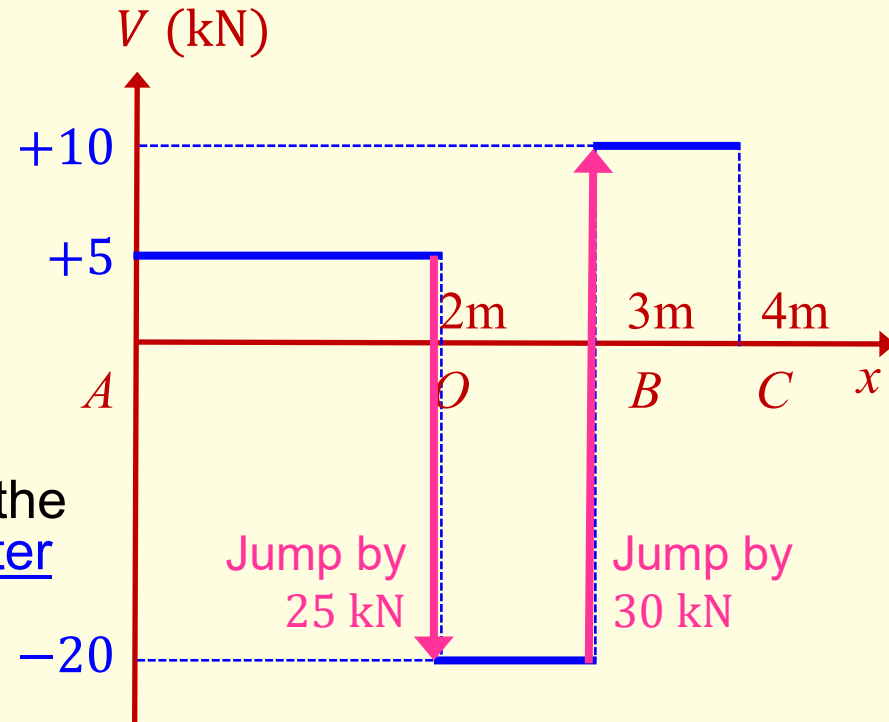
$$\frac{dV}{dx} = w = 0$$

$$\rightarrow V = C_3$$

The constant C_3 can be determined by the “**continuity**” condition at last point *B* after considering the upward jump.

Thus, $C_3 = 10$ kN and

$$V = 10 \text{ kN} \quad (3 \text{ m} \leq x \leq 4 \text{ m})$$



IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram

a) Determine **M** at left end first.

$$x = 0, \quad M = 0$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for the AO region:

$$\frac{dM}{dx} = V = 5 \text{ kN}$$

$$\rightarrow M = 5x + C_4$$

The constant C_4 can be determined by applying the “boundary” condition ($x = 0, M = 0$). Thus, $C_4 = 0$ and **$M = 5x$ ($0 \leq x \leq 2 \text{ m}$)**.

IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram (Cont'd)

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for *OB* region:

$$\frac{dM}{dx} = V = -20 \text{ kN}$$

$$\rightarrow M = -20x + C_5$$

The constant C_5 can be determined by applying the “**continuity**” condition at last point *O* ($x = 2$, $M = 10 \text{ kN} \cdot \text{m}$). Thus, $C_5 = 50 \text{ kN} \cdot \text{m}$ and **$M = -20x + 50$** ($2 \text{ m} \leq x \leq 3 \text{ m}$).

IN-CLASS EXAMPLE, cont'd

Solutions

Step 3: Moment Diagram (Cont'd)

d) Determine **M** diagram according to

$\frac{dM}{dx} = V$ for *BC* region:

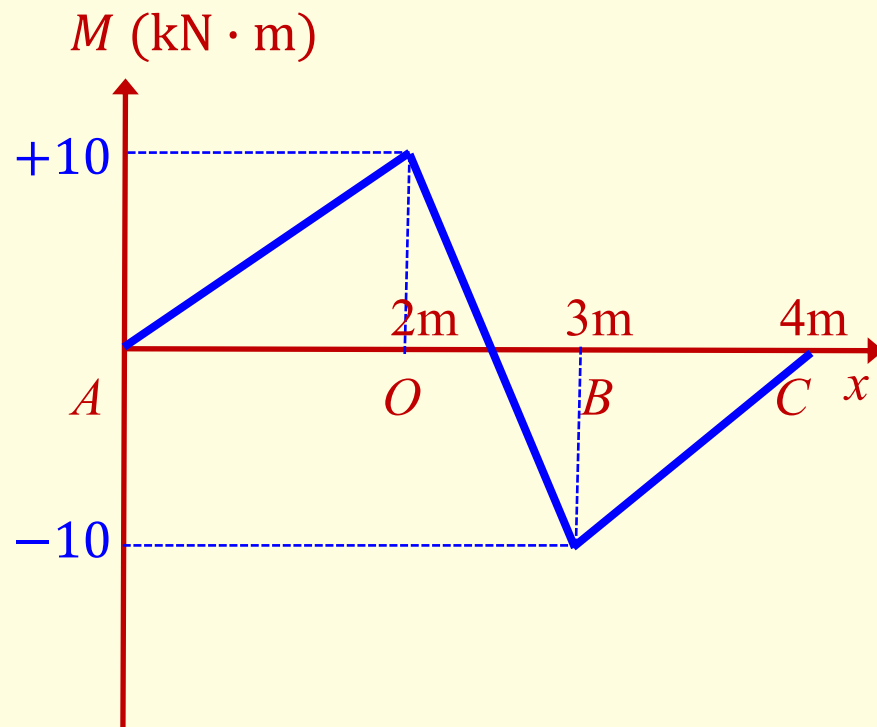
$$\frac{dM}{dx} = V = 10$$

$$\rightarrow M = 10x + C_6$$

The constant C_6 can be determined by applying the “**continuity**” condition at last point *B* ($x = 3$, $M = -10 \text{ kN} \cdot \text{m}$).

Thus, $C_6 = -40 \text{ kN} \cdot \text{m}$ and $M = 10x - 40$ ($3 \text{ m} \leq x \leq 4 \text{ m}$).

Therefore, $M_{\max} = 10 \text{ kN} \cdot \text{m}$ at $x = 2$ & 3 m .



VERIFY: The **M** for at right end *C* ($x = 4 \text{ m}$) is zero, which is well expected!

IN-CLASS EXAMPLE, cont'd

Solutions

Step 4: Section Property

The moment of inertia I of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.05 \times 0.15^3 = 1.40625 \times 10^{-5} \text{ m}^4$$

Step 5: Normal Stress

The maximum normal stress is determined by:

$$\sigma_{\max} = \frac{Mc}{I} = \frac{10000 \times 0.075}{1.40625 \times 10^{-5}} = 53.33 \text{ MPa} \quad \textbf{(Ans.)}$$

RELEVANT PRACTICE

- **Assignment 2 on Blackboard, problem #8 (go to Blackboard under the “Assignments” folder, **Assignment 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**
- **Optional Extra Exercise 2 on Blackboard, problem #6 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 2 due: 8:30 AM, Friday, Oct. 15th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 289 – 300**
 - **In particular, Example 6.11 (pp. 297)**